

Sample Questions based on Lecture 1,2 and 3

1. A company uses an online HR portal where employees must log in using their username and password. Recently, they added fingerprint verification as a second step.

Question:

Which type of security mechanism has the company enhanced?

- A. Authorization
- B. Accounting
- C. Authentication
- D. Access Control

2. Danish Hospital recently started using a new cloud-based patient record system. During an internal audit, the IT security team identified that employees often shared login credentials to speed up data entry — a potential risk of unauthorized data access. The team then evaluated the likelihood and potential impact of this behavior, concluding it could lead to a major data breach and regulatory penalties. To address the issue, management implemented multi-factor authentication (MFA) and conducted staff awareness training. After implementation, periodic audits were scheduled to ensure continued compliance and check if similar risks reappeared.

Question:

Which of the following correctly matches the hospital's actions to the stages of the Risk Management process?

A.

Risk Identification: Sharing login credentials

Risk Analysis/Assessment: Evaluating breach likelihood and impact

Risk Treatment: Implementing MFA and training

Monitoring/Review: Periodic audits

B.

Risk Identification: Implementing MFA

Risk Analysis/Assessment: Checking employee compliance

Risk Treatment: Conducting audits

Monitoring/Review: Identifying login sharing

C.

Risk Identification: Evaluating the impact of a data breach

Risk Analysis/Assessment: Identifying credential sharing

Risk Treatment: Performing risk reviews

Monitoring/Review: Applying MFA

D.

Risk Identification: Conducting audits

Risk Analysis/Assessment: Implementing MFA

Risk Treatment: Evaluating risk likelihood

Monitoring/Review: Identifying login sharing

3. An employee at an organization accidentally shares a confidential file with someone outside the company. The file contains sensitive customer information. The IT team realizes that unauthorized people can now view the data and quickly changes access permissions to stop further sharing.

Question:

What type of threat does this situation represent?

- A. Disruption
- B. Disclosure
- C. Deception
- D. Usurpation

4. A company designs its own encryption system and believes it will remain secure as long as no one outside the company knows how the algorithm works.

However, the encryption key is short and could be guessed easily if the method became public.

What principle of cryptography is the company violating?

- A. The principle that encryption should depend on key secrecy, not algorithm secrecy
- B. The principle of strong password usage
- C. The principle of access control
- D. The principle of data redundancy

5. Which of the following is **NOT** a characteristic of a good cipher?

- A. The amount of secrecy needed should determine the amount of labour appropriate for encryption and decryption.
- B. The set of keys and the enciphering algorithm should be free from unnecessary complexity.
- C. Errors in ciphering should not spread and corrupt the rest of the message.
- D. The size of the ciphertext should be much larger than the original plaintext.

6. Alice wants to send a sensitive message to Bob over an insecure network. She wants to ensure that the message is not altered in transit. Both Alice and Bob share a secret key **K**. Alice computes a MAC using the formula **MAC(K, M) = H(K || M || K)** and sends the message along with the MAC to Bob. Upon receiving it, Bob recalculates the MAC using the shared key and compares it with the one sent.

Question:

What is the primary purpose of using the MAC in this scenario?

- A. To encrypt the message so that no one can read it
- B. To verify the integrity and authenticity of the message
- C. To compress the message for faster transmission
- D. To hide the sender's identity

7. Which of the following statements is correctly matched with its corresponding security property?

- A. Symmetric cryptography — Ensures *non-repudiation* of messages.

- B. Hash functions — Provide *confidentiality* of messages.
 - C. Message Authentication Codes (MAC) — Provide *integrity and authenticity* of a message using a shared secret.
 - D. Asymmetric cryptography — Requires *both parties* to know the same secret key.
8. A self-enforcing protocol is best described as:
- A. A protocol that relies on a trusted third party to verify all actions
 - B. A protocol that enforces fairness and correctness without external help
 - C. A protocol dependent on manual verification
 - D. A protocol requiring judicial review
9. A research organization called Security Labs is working on a confidential drone project. Only two engineers, Alice and Bob, are allowed to share encrypted design files. They use a symmetric key cryptography protocol to secure their communications. Before starting their collaboration, Alice and Bob need to agree on a secret key for AES encryption. They decide to send USB through third party courier which contain a USB drive containing the shared key K_{AB} . What security issue is most relevant in this phase?
- A. Key reuse
 - B. Secure key distribution
 - C. Ciphertext expansion
 - D. Hash collisions
10. Which statement is correct about public key cryptography?
- A. Both sender and receiver use the same key for encryption and decryption.
 - B. The private key is shared publicly, while the public key is kept secret.
 - C. It uses a pair of mathematically related keys — one public and one private.
 - D. It is faster and more efficient than symmetric key cryptography for bulk data encryption.